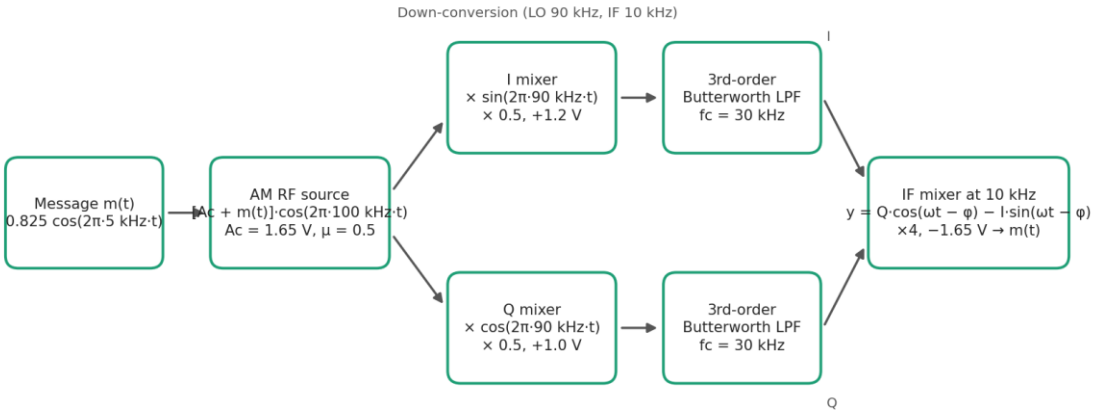
01. Project Overview: Low-IF AM Receiver with a Transistor-Level Analog Baseband

1. Main idea

This project simulates, in LTspice, the analog baseband of a **low-IF receiver** that recovers a 5 kHz message from an amplitude-modulated (AM) carrier. The RF signal and the local oscillators are ideal behavioral sources. Everything after the mixers is built from real circuits on a single 3.3 V supply:

1. **I/Q down-conversion.** The RF signal is multiplied by two LO signals 90 degrees apart (sine and cosine) to produce an in-phase (I) and a quadrature (Q) signal at a low intermediate frequency (IF).
2. **Channel-select / anti-alias filtering.** A **3rd-order Butterworth active-RC low-pass filter** (30 kHz cutoff) removes the sum-frequency product and passes the IF. The filter is built from a **two-stage CMOS Miller op-amp** designed at transistor level.
3. **Second quadrature mixing at the IF.** I and Q are combined with a 10 kHz quadrature pair so the IF carrier is removed and the original message is reconstructed.

All frequencies are scaled down (100 kHz carrier instead of MHz) so transient simulations stay fast, while the architecture is the same as in a real low-IF receiver.



System block diagram

2. Message signal

A single-tone message is used:

$$m(t) = A_m \cos(2\pi f_m t), \qquad A_m = 0.825 \text{ V}, \quad f_m = 5 \text{ kHz}$$

In the LTspice schematic it is the source `V1 = SINE(0 0.825 5k 0 0 90 100)` (a 90 degree phase makes it a cosine). It is used only as the reference to compare against the recovered signal.

3. Modulation: how the RF signal is generated

The RF signal is **conventional AM (double-sideband with carrier)**:

$$s(t) = \left[A_c + m(t)\right] \cos(2\pi f_c t) = A_c\left[1 + \mu \cos(2\pi f_m t)\right] \cos(2\pi f_c t)$$

Symbol	Meaning	Value

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A_c	carrier amplitude	1.65 V
f_c	carrier frequency	100 kHz
$\mu = A_m / A_c$	modulation index	0.5 (50 %)

In LTspice (source B3): `V = 1.65*(1+0.5*cos(2*pi*5k*time))*cos(2*pi*100k*time)`.

The envelope swings between $A_c(1 - \mu) = 0.825$ V and $A_c(1 + \mu) = 2.475$ V. Expanding the product shows the spectrum: a carrier at 100 kHz and two sidebands, each of amplitude $A_m/2 = 0.4125$ V:

$$s(t) = A_c \cos \omega_c t + \tfrac{A_m}{2} \cos(\omega_c + \omega_m)t + \tfrac{A_m}{2} \cos(\omega_c - \omega_m)t$$

The sidebands sit at 95 kHz and 105 kHz, so the RF bandwidth is $2f_m = 10$ kHz.

4. Reconstruction: how the message is recovered

4.1 I/Q mixers

The local oscillator is at $f_{LO} = 90$ kHz, so the IF is $f_{IF} = f_c - f_{LO} = 10$ kHz. Two LO phases are used:

$$v_{LO,I} = \sin(\omega_L t), \qquad v_{LO,Q} = \cos(\omega_L t)$$

Each mixer multiplies the RF by an LO and scales by $k = 0.5$ (the behavioral source `V(rf)*V(lo)*0.5 + offset`). Using the product-to-sum identities:

$$k\, s(t) \sin \omega_L t = \tfrac{k}{2} \left[A_c + m\right] \left[\sin(\omega_c + \omega_L)t - \sin(\omega_c - \omega_L)t\right]$$

$$k\, s(t) \cos \omega_L t = \tfrac{k}{2} \left[A_c + m\right] \left[\cos(\omega_c - \omega_L)t + \cos(\omega_c + \omega_L)t\right]$$

Each output has a **difference term at the 10 kHz IF** (wanted) and a **sum term at 190 kHz** (unwanted).

4.2 Low-pass filtering

The 3rd-order Butterworth filter (cutoff $f_{c,LPF} = 30$ kHz) has

$$|H(f)| = \frac{1}{\sqrt{1 + (f/f_{c,LPF})^6}}$$

so it passes the 10 kHz IF (gain 0.999, about 0 dB) and attenuates the 190 kHz sum term by $1/\sqrt{1 + (190/30)^6} \approx 0.004$ (about -48 dB). After filtering, with $\omega_{IF} = \omega_c - \omega_L$ and $k/2 = 0.25$:

$$I(t) = -\tfrac{k}{2} \left[A_c + m(t)\right] \sin(\omega_{IF} t), \qquad Q(t) = +\tfrac{k}{2} \left[A_c + m(t)\right] \cos(\omega_{IF} t)$$

The message is now an amplitude envelope on a 10 kHz carrier, shared between I and Q.

4.3 Second quadrature mixer at the IF

Combining I and Q with a quadrature oscillator at exactly the IF removes the IF carrier:

$$y(t) = Q(t) \cos(\omega_{IF} t) - I(t) \sin(\omega_{IF} t) = \tfrac{k}{2} \left[A_c + m(t)\right] \left(\cos^2 + \sin^2\right) = 0.25 \left[A_c + m(t)\right]$$

The double-frequency terms cancel exactly, so no extra filter is needed.

Phase compensation. The low-pass filter delays the 10 kHz IF in both I and Q by a phase φ_f. For a 3rd-order Butterworth at $x = f/f_c = 1/3$:

$$\varphi_f = \arctan(x) + \arctan\left(\frac{x}{1-x^2}\right) = 18.4^\circ + 20.6^\circ = 39^\circ \approx 0.68 \text{ rad}$$

If the second oscillator used zero phase, the output would shrink by cos(39°) = 0.78. So the oscillator is delayed by the same phase: cos($\omega_{IF}t$ − 0.68) and sin($\omega_{IF}t$ − 0.68). Then y = 0.25 [A_c + $m(t)$] exactly.

Scaling. Multiplying by 4 cancels the mixer gain and the 1/2 from the trigonometric identity, giving A_c + $m(t)$. Subtracting A_c = 1.65 V removes the carrier term and leaves the message:

$$v_{mre}(t) = 4\,y(t) - 1.65 = m(t)$$

LTspice (source B4): `V=4*((V(vqlpf)-1)*cos(2*pi*10k*time-0.68)-(V(vilpf)-1.2)*sin(2*pi*10k*time-0.68))-1.65`

4.4 DC levels for a single 3.3 V supply

The op-amps only work between about 0.4 V and 2.9 V, so every signal is centered at a safe DC level:

- The filters reference their non-inverting inputs to V_{ref} = 1.65 V (half the supply).
- The I mixer adds +1.2 V and the Q mixer adds +1.0 V to center their outputs. The 3rd-order filter has DC gain +1 (two inverting sections), so these levels pass straight through.
- The `-1.2` and `-1` in B4 remove these offsets before the second mixing.

5. Specifications

Parameter	Value
Supply	3.3 V, single supply
Message	5 kHz tone, 0.825 V peak
RF signal	AM with carrier, f_c = 100 kHz, A_c = 1.65 V, μ = 0.5
Local oscillator	90 kHz, 1 V peak, quadrature (sine and cosine)
IF	10 kHz
Mixer gain / offsets	0.5, I offset 1.2 V, Q offset 1.0 V
Low-pass filter	3rd-order Butterworth, active-RC, f_c = 30 kHz, DC gain 1, 60 dB/decade roll-off
Op-amp	2-stage CMOS with Miller compensation, DC gain about 80 dB, 405 µA supply current each
Reconstruction	10 kHz quadrature mixer, phase 0.68 rad, gain 4, DC removal 1.65 V
Recovered message	about 0.83 V peak, 5 kHz, delayed by about 10 µs (filter group delay)

6. Limitations and possible extensions

- **Ideal sources.** The RF signal, the LOs and the mixers are behavioral sources. A transistor-level mixer (for example Gilbert cell) and LO generation are not included.
- **Carrier/LO synchronization.** The 10 kHz second oscillator is exactly the carrier-minus-LO difference with a known phase by construction. A real receiver needs carrier recovery (PLL or Costas loop).
- **Image frequency.** A signal at 80 kHz (the image of 100 kHz about the 90 kHz LO) is not removed by the real low-pass filter. With this second mixer it is moved to 20 kHz, away from the message band, but still needs a baseband filter (or a complex I/Q filter) to be rejected.
- **Op-amp phase margin.** With Cc = 0.1 pF the stand-alone op-amp has a low unity-gain phase margin (see 03_circuits_and_design.docx). Operating inside the filter feedback loops it is acceptable, and no ringing is seen in the transient results, but 0.5 to 1 pF would give more margin.
- **DSB-SC.** The same demodulator works for DSB-SC (no carrier) with gain 2 and no DC subtraction. This was checked with a Python model of the chain, not in LTspice.